Section L
Antiderivatives

1. $f(x) = \int x^8 dx = \frac{1}{9}x^9 + c$

2. $f(x) = \int x^{-6} dx = -\frac{1}{5}x^{-5} + C$

3. $f(x) = \int x^{5/7} dx = \frac{7}{12}x^{\frac{12}{7}} + c$

4. $f(x) = \int \frac{4}{\sqrt{x}} dx = (4)(2)\sqrt{x} + c$

5. $f(x) = \int \frac{1}{2x^3} dx = \left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)x^{-2} + c$

6. $f(x) = \int x^3\sqrt{x}\, dx = \int x^{\frac{7}{2}} dx$
$= \frac{2}{9}x^{\frac{9}{2}} + c$

7. $f(x) = \int \sqrt[3]{t^2}\, dt = \int t^{\frac{2}{3}} dt$
$= \frac{3}{5}t^{\frac{5}{3}} + C$

8. $f(x) = \int (x^3 - 2x + 7)\, dx$
$= \frac{1}{4}x^4 - (2)\left(\frac{1}{2}\right)x^2 + 7x + C$

9. $f(x) = \int \left(x^{-3} + x^{\frac{1}{2}} - 3x^{\frac{1}{4}} + x^2\right) dx$
$= -\frac{1}{2}x^{-2} + \frac{2}{3}x^{3/2} - (3)\left(\frac{4}{5}\right)x^{\frac{5}{4}} + \frac{1}{3}x^3 + C$

10. $f(u) = \int \left(u^{\frac{2}{3}} - 4u^{\frac{1}{5}} + 4\right) dw$
$= \frac{3}{5}u^{\frac{5}{3}} - (4)\left(\frac{5}{6}\right)u^{\frac{6}{5}} + 4w + C$

11. $f(x) = \int \left(\frac{7}{x^{3/4}} - \sqrt[3]{x} + 4\sqrt{x}\right) dx$
$= (7)(4)x^{\frac{1}{4}} - \frac{3}{4}x^{\frac{4}{3}} + (4)\left(\frac{2}{3}\right)x^{3/2} + C$

12. $f(x) = \int (x + x^4)\, dx$
$= \frac{1}{2}x^2 + \frac{1}{5}x^5 + c$

13. $f(t) = \int (2 - t + 2t^2 - t^3)\, dt$
$= 2t - \frac{1}{2}t^2 + (2)\left(\frac{1}{3}\right)t^3 - \frac{1}{4}t^4 + c$

14. $f(x) = \int x^{\frac{1}{3}}(4 - 4x + x^2)\, dx$
$= \int \left(4x^{1/3} - 4x^{\frac{4}{3}} + x^{\frac{7}{3}}\right) dx$
$= (4)\left(\frac{3}{4}\right)x^{\frac{4}{3}} - (4)\left(\frac{3}{7}\right)x^{\frac{7}{3}} + \left(\frac{3}{10}\right)x^{\frac{10}{3}} + C$

15. $f(x) = \int \left(\frac{1}{x^2} - \cos x\right) dx$
$= -x^{-1} - \sin x + C$

16. $f(\theta) = \int (4\sin\theta + 2\cos\theta)\, d\theta$
$= -4\cos\theta + 2\sin\theta + c$

17. $f(y) = \int e^y dy = e^y + c$

18. $f(x) = \int x e^{x^2} dx$
$= \frac{1}{2}e^{x^2} + c$

19. $f(x) = \int \frac{2}{x} dx$
$= 2\ln|x| + c$